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# **Twin Synergy Telecommunications**

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## **Network Infrastructure**

Sol in Libra, 2026 e.v.  
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Keeper of the Secret Treasure

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# Network Infrastructure — Ground-Up Blueprint, Honest Phases

"Their line is gone out through all the earth, and their words to the end of the world."

— [Tanakh: Psalm 19:4, KJV]

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## Status Marks & Phase Legend

Every claim in this document carries a status mark. No claim floats unmarked.

- **[VERIFIED FACT]** — checked against a live or primary source cited in place.
- **[DESIGN]** — an engineering decision of this blueprint; it stands until his red pen rules otherwise.
- **[HYPOTHESIS]** — an assumption awaiting verification; the document names where to verify it.
- **[SCRIBE]** — scribe's inference or arithmetic; honest working shown, not doctrine.

Phases:

- **PHASE 1 — BUILDABLE NOW.** Resale and integration on the Starlink business pathway, leased transit, first Point of Presence (PoP), DNS infrastructure, no plant built.
- **PHASE 2 — OWN PLANT.** Ground-station partnerships, additional PoPs, owned backbone segments, the ICANN registrar accreditation path.
- **PHASE 3 — ASPIRATION.** Marked as such throughout: deeper constellation integration or own LEO capacity, owned subsea/terrestrial backbone.

This is a design and blueprint document. It is **not** a legal filing and claims **no** accreditation, license, or regulatory status **[DESIGN]**.

## Scientific Illuminism Framing — The Build as Experiment

The method of science, the aim of religion. The whole network is one experiment:

- **HYPOTHESIS:** A sovereign communications carrier can be built from the ground up on the Starlink business pathway — resale and integration first, owned plant later — without surrendering the five doctrines (fail-closed, least privilege, honest metrics, IPv6-first, privacy/no-logging), and without ever claiming capacity it does not hold.
- **METHOD:** Phase the build so each phase pays for and proves the next. Phase 1 rents the road and operates the edge honestly; Phase 2 lays owned road where economics permit; Phase 3 is aspiration, never promise. Every specification below is marked to its evidence tier.

- **OBSERVATION:** To be recorded in the honest-metrics publication (§7) — measured medians and p95s, not "up to" theater; outages named and dated, not smoothed.
  - **RESULT:** To be determined by his red pen and by the field. This document is the apparatus, not the result.
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## §1. Doctrine — The Five Non-Negotiables

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[DESIGN] These govern every section below. They are load-bearing, not decorative.

- **Fail-closed.** When a component cannot verify, it refuses — loudly. No silent degradation into an insecure state. Silence is a lie; refusal is truth.
- **Least privilege.** Every service holds the minimum authority to do its job. The resolver does not see the customer's payload; the edge router does not see the customer's DNS; the billing system does not see either.
- **Honest metrics.** Publish measured speeds (median, p95), measured uptime, measured latency — never "up to" speeds, never unqualified uptime. If a number cannot be measured, it is not published; it is marked [HYPOTHESIS].
- **IPv6-first thinking.** Dual-stack everywhere, but the design assumes IPv6 is the native protocol and IPv4 is the legacy compatibility layer — because that is the direction of the network.
- **Privacy (no-logging resolver).** The recursive resolver keeps no query logs. Not "anonymized" logs — *no* logs. What is never recorded cannot be subpoenaed, breached, or sold.

And the constitutional layer beneath all five, §9: **common carriage** — no DPI, no throttling, no paid fast lanes. The carrier carries; it does not judge, inspect, or favor.

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## §2. Upstream & Transit

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### PHASE 1 — Starlink Business as the pathway (buildable now)

**HYPOTHESIS:** SpaceX operates authorized-reseller and enterprise-integration channels for Starlink Business that a new carrier can enter — either as an authorized reseller or as an enterprise integrator managing fleets of Business terminals for its customers. **Verify with:** SpaceX enterprise sales ([starlink.com/business](https://starlink.com/business)) and any published authorized-reseller program terms [HYPOTHESIS].

The design does not depend on the reseller badge. Two doors, one house [DESIGN]:

- **Door A — Authorized reseller.** Twin Synergy sells and provisions Starlink Business service under SpaceX's program, adds managed edge (router, DNS, support) as its own value layer.
- **Door B — Enterprise integration.** Twin Synergy operates as the customer's agent: the customer holds the Starlink Business account; Twin Synergy provisions, manages, monitors, and supports the deployment — edge routers, DNS, failover — as a managed service.

Either door opens Phase 1. Both are honest: the document never claims Twin Synergy is the satellite operator.

### **Starlink Business pathway specs — approximate, subject to SpaceX's current terms**

**[SCRIBE]:** Third-party reporting (verified 2026-10-09 against public sources; **re-verify at starlink.com before committing**) indicates:

- Priority data buckets (business "Priority" plans): reported tiers of 40 GB / 1 TB / 2 TB monthly priority data, with unlimited standard data thereafter; overage charges reported around \$0.50-\$1.00/GB depending on plan vintage **\*\*[SCRIBE — re-verify]\*\***.
- Published speeds on the Priority critical-information summary: download 120-270 Mbps, upload 12-35 Mbps, latency 25-60 ms, with the explicit statement that stated speeds and uninterrupted use are *\*not guaranteed\** **\*\*[SCRIBE — re-verify; note the honesty of the source: SpaceX itself disclaims the guarantee, and so shall we]\*\***.
- Flat High-Performance terminal required for business-grade service; hardware cost reported in the ~\$1,999-\$2,500 range **\*\*[SCRIBE — re-verify]\*\***.
- Priority plans include a publicly routable IPv4 address **\*\*[SCRIBE — re-verify]\*\***.

The blueprint takes SpaceX at its word: speeds are published ranges, never guarantees. Our honest-metrics publication (§7) will publish our own measured ranges and say so.

**Leased backbone/transit (Phase 1) [DESIGN]:** From day one the company holds its own AS number and address space, peers at the first PoP, and buys IP transit from at least **two** independent Tier-1/2 transit providers for redundancy. Rationale: the upstream is leased, but the routing policy is ours — multi-homed from the start, so no single transit failure is fatal. Commit to RPKI route-origin validation on all sessions from day one (Phase-1 buildable; see §5 and §9).

### **PHASE 2 — Owned segments**

- Ground-station partnerships: capacity agreements with teleport operators rather than building dishes — partnership, not concrete **\*\*[DESIGN]\*\***.
- Additional PoPs in a second and third metro (site selection by customer density and transit diversity, not prestige).
- Owned backbone segments where the economics clear: metro fiber leases with IRU (indefeasible right of use) terms, then dark fiber where traffic justifies it **\*\*[DESIGN]\*\***.
- Registrar accreditation path: the real requirements, honestly stated — §10.

### **PHASE 3 — ASPIRATION**

Deeper constellation integration or own LEO capacity; owned subsea or long-haul terrestrial backbone. **This is aspiration, not a plan.** No dates, no budgets, no promises. It is written here so the ambition is on record and the honesty is on record beside it **[DESIGN]**.

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## **§3. PoP Architecture**

### **PHASE 1 — First PoP [DESIGN]**

One PoP, properly built, beats three PoPs sketched on a napkin.

- **\*\*Site:\*\*** a carrier-neutral colocation facility in a metro with  $\geq 3$  independent transit options and an Internet exchange (IX) presence. Selection criteria published; the choice is engineering, not branding.
- **\*\*Contents:\*\*** redundant edge routers (two, active/passive with VRRP or active/active with ECMP), the authoritative DNS anycast node (§5), the recursive resolver cluster (§5), monitoring/metrics collectors (§7), and the management jump host — loopback-only management plane, no public SSH, keys only, per the fail-closed doctrine.
- **\*\*Upstream:\*\*** the two leased transit sessions plus IX peering. Starlink Business terminals at customer sites ride SpaceX's ground infrastructure; the PoP is where Twin Synergy's *\*own\** routing policy, DNS, and monitoring live.
- **\*\*Power/cooling:\*\*** N+1 on power feeds; the PoP is not declared "high availability" until the failover has been *\*tested\** — a failover drill is part of commissioning, and the drill's result is published (§7).

## PHASE 2 — PoP 2 and PoP 3 [DESIGN]

Second metro chosen for transit diversity (different upstream providers where possible); anycast DNS gains true geographic diversity; resolver clusters follow. No PoP is added until PoP 1's honest metrics clear the bar the company publishes.

## PHASE 3 — ASPIRATION

Edge PoPs co-located with ground-station partnerships; the map grows only as the measurements justify it.

## §4. Last-Mile Options

### Starlink terminal tiers (Phase 1) [DESIGN]

| Tier              | Terminal (reported)                                   | Use                                             | Status                                             |
|-------------------|-------------------------------------------------------|-------------------------------------------------|----------------------------------------------------|
| Business Priority | Flat High-Performance (~\$1,999-\$2,500 hw, reported) | Primary customer edge; managed router behind it | [SCRIBE — re-verify pricing/specs at starlink.com] |
| Standard / Mini   | Standard or Mini kit                                  | Backup path, portable, low-data sites           | [SCRIBE — re-verify]                               |

The managed router is Twin Synergy's product at the edge: customer LAN, dual-WAN failover where a second path exists (cellular backup, second terminal), DNS pointed at the no-logging resolver, IPv6 enabled by default, remote management over an encrypted channel the customer can audit [DESIGN].

### Fixed wireless (Phase 1 where economical) [DESIGN]

Where a customer site has line-of-sight to a viable fixed-wireless provider, offer it as primary or backup. The doctrine does not bless technologies; it blesses measurements. Fixed wireless is offered only where measured latency/loss clear the published bar.

### Fiber where economical (Phase 1→2) [DESIGN]

Where fiber is available at the customer site at sane cost, it is the preferred last mile — lowest latency, highest honesty. The company does not overbuild fiber in Phase 1; it buys it where it exists and builds toward ownership only in Phase 2 where traffic justifies IRUs.

**Honest publication rule:** every last-mile option ships with its measured profile (median/p95 down/up, latency, loss, observed outage minutes per month) — and the profile is per-technology, never blended into a single "network average" that hides the weak path **[DESIGN]**.

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## §5. DNS Infrastructure

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### Authoritative DNS — anycast **[DESIGN]**

- **\*\*Phase 1:\*\*** anycast announcement of the authoritative name-server addresses from the first PoP via the two transit providers; zone data replicated with signed transfers (TSIG) between hidden primary and the public anycast nodes. Honest note: with one PoP, anycast buys routing resilience, not geographic diversity — the document says so plainly.
- **\*\*Phase 2:\*\*** anycast nodes at PoP 2 and PoP 3; true geographic diversity.
- **\*\*DNSSEC throughout:\*\*** every zone Twin Synergy is authoritative for is DNSSEC-signed from Phase 1 (algorithm: ECDSAP256SHA256 or Ed25519 per current best practice at signing time; key ceremonies documented, KSK/ZSK split, emergency rollover procedure written *\*before\** it is needed). Unsigned zones are a finding, not a footnote **\*\*[DESIGN]\*\***.

### Recursive resolver — the no-logging doctrine **[DESIGN]**

- Offered to all customers as the default resolver on the managed edge router.
- **\*\*No query logs.\*\*** Not anonymized, not sampled, not "retained 24 hours" — none. The resolver is configured to discard query data after answering. What is never recorded cannot be subpoenaed, breached, or sold (the doctrine, §1).
- **\*\*QNAME minimization\*\*** enabled (RFC 9156): the resolver sends the minimum necessary labels upstream — a privacy feature that costs nothing and leaks less **\*\*[VERIFIED FACT: RFC 9156 standardizes QNAME minimization]\*\***.
- **\*\*DNS-over-TLS and DNS-over-HTTPS\*\*** offered on the standard ports; opportunistic and strict profiles documented.
- **\*\*DNSSEC validation\*\*** enabled on the resolver from Phase 1 — bogus answers are refused (fail-closed: SERVFAIL, never a lie) **\*\*[DESIGN]\*\***.
- Aggressive negative caching (RFC 8198) to reduce upstream chatter **\*\*[VERIFIED FACT: RFC 8198 standardizes aggressive use of DNSSEC-validated cache]\*\***.
- **\*\*Transparency:\*\*** the resolver's configuration (minus secrets) is published so the no-logging claim is checkable. A privacy claim that cannot be checked is a marketing claim; this one is engineered to be checkable **\*\*[DESIGN]\*\***.

### Registrar-side DNS (Phase 2)

When accreditation is achieved (§10), every registered domain gets DNSSEC-capable delegation by default, with secDNS EPP support and a one-click signing flow. The registrar never holds customer

DNS hostage: zone export is always available, transfers out are never throttled or obstructed **[DESIGN]**.

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## §6. IPv6 Plan — IPv6-First Dual-Stack

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**[DESIGN]** The network is designed IPv6-first: every new service is built on IPv6 and then given IPv4 compatibility — never the reverse.

- **\*\*Address space:\*\*** obtain an IPv6 /32 (or larger per need) from ARIN under the NRPM for ISP allocations **\*\*[HYPOTHESIS — verify current ARIN NRPM ISP allocation policy and fees at arin.net before applying]\*\***; obtain an Autonomous System Number at the same time.
  - **\*\*Customers:\*\*** each customer site receives an IPv6 /56 (or /48 on justified request) — enough subnets for a site to grow without renumbering. IPv4 via the publicly-routable address on the Starlink Business path plus CGNAT only where unavoidable, and CGNAT is disclosed, never hidden **\*\*[DESIGN]\*\***.
  - **\*\*Starlink IPv6:\*\*** Starlink is reported to delegate IPv6 prefixes to customer routers **\*\*[SCRIBE — verify current delegation size and behavior on the live service]\*\***; the managed edge router is configured to accept the delegation and advertise it downstream, preferring it over any tunnel.
  - **\*\*DNS:\*\*** all authoritative nameservers and the resolver reachable over IPv6 from Phase 1; AAAA records published alongside A records; the company's own web presence dual-stack.
  - **\*\*Measurement:\*\*** the honest-metrics publication (§7) reports IPv6 vs IPv4 traffic share — the IPv6-first claim is measured, not asserted **\*\*[DESIGN]\*\***.
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## §7. Monitoring & Honest Metrics

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The anti-weasel-word section. **[DESIGN]**

**What is published (public status page, updated continuously):**

- Measured speed per last-mile technology: median and p95 download/upload, updated daily from active probes — never "up to," never a theoretical maximum.
- Latency (median/p95) and packet loss per path, per PoP.
- Uptime per service (resolver, authoritative DNS, PoP transit) as *\*measured availability with the outage log attached\** — every incident named, dated, and given a cause. "99.9% uptime" without the incident log is a weasel number; the log is the number.
- IPv6 traffic share (§6).
- Failover drill results (§3): date, what was failed over, whether it worked.

**What is never published:** "up to" speeds, unqualified uptime percentages, blended averages that hide a weak path, or any number that cannot be reproduced from the published methodology **[DESIGN]**.

**Methodology transparency:** probe design, sample sizes, and aggregation methods are published beside the numbers. A metric whose method is secret is an advertisement **[DESIGN]**.

**Internal monitoring:** per-customer edge telemetry (with customer-visible dashboards — the customer sees what we see about their own line; least privilege means their data is theirs), PoP health, BGP session state, DNS query rates (never query contents — the no-logging doctrine binds monitoring too), certificate expiries, and RPKI validation state **[DESIGN]**.

## §8. Failure Modes — Fail-Closed Behavior

For each failure, the specified behavior. "Fail-closed" means: refuse loudly, never degrade silently into an insecure or dishonest state **[DESIGN]**.

| Failure                                                              | Behavior                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Upstream transit loss (one of two)                                   | Traffic converges on the surviving transit; alert fires; status page notes degraded redundancy. Single-homed is a declared emergency state, not a quiet one.                                                                                                                                                              |
| Upstream transit loss (both) / Starlink path loss at a customer site | Customer edge stops forwarding *new* sessions rather than hairpinning through untrusted or unmeasured paths. Existing sessions drain. The edge serves a local status notice (captive notice page on HTTP, ICMP unreachable otherwise) stating the outage — never a silent black hole, never a fake "connected" indicator. |
| DNSSEC validation failure (bogus signature)                          | SERVFAIL. The resolver returns no answer rather than a forged one. Logged as an event (rate only), surfaced on the status page.                                                                                                                                                                                           |
| Authoritative node loss                                              | Anycast withdraws the failed node; remaining nodes serve. If *all* nodes fail, zones go dark rather than serving stale unsigned data — dark is honest, stale is a lie.                                                                                                                                                    |
| Management plane unreachable                                         | No remote changes accepted until the management channel is re-verified. The network keeps forwarding on last-known-good config; it does not accept new config over an unverified channel.                                                                                                                                 |
| Power loss at PoP                                                    | Graceful shutdown order: resolver, authoritative, then routing — so DNS answers stop before routing lies about reachability. On restore, services verify (DNSSEC chain, RPKI, BGP) *before* announcing.                                                                                                                   |
| Certificate expiry (DoT/DoH, status page)                            | Monitoring alerts at 30/14/7 days; expiry is treated as a P1 incident because an expired certificate trains users to click through warnings — and that is a security failure, not a cosmetic one.                                                                                                                         |

**The rule beneath the table:** every failure mode has a specified behavior written before it happens. An unspecified failure mode is a promise to improvise, and improvisation under pressure is where dishonest systems are born **[DESIGN]**.

## §9. Security Posture — No DPI, No Throttling, Constitutional Common Carriage

**[DESIGN]** The carrier carries. It does not inspect, judge, or favor.

- **\*\*No deep packet inspection.\*\*** The network forwards packets; it does not open them. Traffic management, if ever needed for genuine congestion, is protocol-agnostic and disclosed — never content-based, never customer-based.
- **\*\*No throttling.\*\*** No paid fast lanes, no application-specific degradation, no "network management" that coincidentally punishes competitors. Congestion is met with capacity or with honest, disclosed, content-neutral queuing — and the queuing policy is published.
- **\*\*Common carriage as constitutional posture:\*\*** the company's doctrine holds that a carrier

offering its services to all who pay its charges is a common carrier in the Black's sense — "one who, by virtue of his calling and as a regular business, undertakes for hire to transport persons or commodities from place to place, offering his services to all such as may choose to employ him and pay his charges"\* [Black's Law Dictionary, 2d ed. (1910), definition of CARRIER, citing *Iron Works v. Hurlbut*, 158 N.Y. 34, 52 N.E. 665, 70 Am. St. Rep. 432] **\*\*[VERIFIED FACT — read verbatim from the 1910 text in this workspace]\*\***. Packets are the commodities; the principle is the same. The carrier does not get a vote on the cargo.

- **\*\*Law footnote:\*\*** unjust or unreasonable discrimination by communications common carriers is unlawful under [Law: 47 U.S.C. § 202(a)] — cited as reference, not as legal advice and not as a claim of legal effect **\*\*[VERIFIED FACT — statute exists as cited]\*\***.
  - **\*\*Encryption posture:\*\*** TLS everywhere on company surfaces; DNS-over-TLS/HTTPS for resolver privacy (§5); no TLS interception, no "helpful" middleboxes that break end-to-end encryption — a middlebox that decrypts customer traffic is DPI with better branding, and it is refused.
  - **\*\*RPKI:\*\*** route-origin validation on all BGP sessions from Phase 1; ROAs published for all company prefixes; invalids dropped loudly (fail-closed routing) **\*\*[DESIGN]\*\***.
  - **\*\*Abuse handling:\*\*** standard, published abuse contact (RFC 2142 abuse@ role account); action on verified abuse reports (spam, phishing, C2) through suspension of the \*account\*, never through silent traffic manipulation — and every action appealable with reasons given in writing **\*\*[DESIGN]\*\***.
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## **§10. Registrar Accreditation Path (Phase 2) — The Real Requirements, Honestly Stated**

The company intends to become an ICANN-accredited registrar in Phase 2. The requirements below are stated as ICANN states them — no softening, no "streamlined" language **[VERIFIED FACT — per ICANN's published Registrar Accreditation Application instructions and Statement of Registrar Accreditation Policy, [icann.org](https://www.icann.org/policy)]**:

- **\*\*Liquid working capital:\*\*** USD \$70,000 deemed sufficient at the commencement of the accreditation period — demonstrated by bank statement, guaranteed bank loan certificate, or letter of credit from a recognized financial institution (or audited financials for existing businesses).
- **\*\*Commercial general liability insurance:\*\*** policy limit of USD \$500,000 deemed sufficient; certificate required as a condition of accreditation becoming effective.
- **\*\*Application fee:\*\*** USD \$3,500, non-refundable.
- **\*\*Annual accreditation fee:\*\*** USD \$4,000 per year.
- **\*\*The application itself:\*\*** ~58 questions across ~25 pages covering business model, technical capability (EPP — Extensible Provisioning Protocol — operated in-house or via a compliant provider), and compliance posture.
- **\*\*Background checks:\*\*** third-party verification of directors, officers, and 5%+ owners; unresolved legal issues are disqualifying facts, not paperwork.

- **The RAA:** signature of the Registrar Accreditation Agreement — full compliance with its terms (data escrow, WHOIS/RDAP accuracy obligations, abuse handling, registrant protections) is the ongoing price of the accreditation.
- **Operational domain:** hold an existing, operational second-level domain at the time of application.

**Phase-1 posture:** operate as a reseller under an accredited registrar while building the technical plant (EPP client, DNSSEC provisioning, escrow-ready data handling) so the accreditation application, when filed, describes a running system rather than a plan **[DESIGN]**. The document claims no accreditation until ICANN grants it.

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## §11. Four Wells Audit Brief

Per the Symmetrical Directive of the Four Wells — each well must actually say the thing.

- **Law:** the common-carriage posture (§9) is grounded in the Black's 2d-ed. definition of the common carrier, quoted verbatim above, and footnoted to [Law: 47 U.S.C. § 202(a)] as reference. The ICANN requirements (§10) are stated from ICANN's own published policy, unsoftened. Genuine.
  - **Religion (Scripture):** the epigraph — "Their line is gone out through all the earth, and their words to the end of the world" [Tanakh: Psalm 19:4, KJV] — is the psalmist describing a transmission that reaches everywhere; it is what a telecommunications carrier is built to do. Companion: "We made you peoples and tribes, that you may know one another" [Qur'an: al-Ḥujurāt 49:13, Pickthall] — the network as the instrument of mutual knowing. Genuine; neither is forced.
  - **Hip-hop:** \*well held open this pass.\* No correlation found that meets the bar — no forced lyric, no fabricated link. The red-pen desk holds this well for his ruling or a future pass.
  - **Fiction & film:** "The sky above the port was the color of television, tuned to a dead channel." [William Gibson, *Neuromancer*, 1984] — the opening line of the novel that named cyberspace, describing the sky itself as a tuned medium. A carrier that rides a satellite constellation is, literally, in the business of the tuned sky. Genuine.
  - **Humanity:** the design serves the unserved first — the rural site, the curbside worker, the place the fiber map forgot. The last-mile table (§4) leads with satellite precisely because the profitable places already have carriers and the forgotten places do not. Connectivity is not charity here; it is the business model pointed at the people the incumbents priced out. Genuine.
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## §12. Red-Pen Questions Held Open

For his ruling — named once, not nagged:

- **Door A or Door B first (§2):** pursue authorized-reseller status with SpaceX, or open as an enterprise integrator on customer-held Business accounts? The blueprint supports either; the business posture differs.

- **\*\*First PoP metro:\*\*** engineering criteria are specified (§3); the actual city is his call.
  - **\*\*Fixed-wireless build vs. buy:\*\*** partner with existing WISPs in Phase 1, or hold fixed wireless for Phase 2?
  - **\*\*The hip-hop well (§11):\*\*** held open for his red pen — a genuine correlation or a standing thin well by his ruling.
  - **\*\*Phase-2 timing trigger:\*\*** what measured threshold (customer count, revenue, traffic) authorizes the accreditation filing and PoP 2 spend?
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